TEVORA™

Telco Cloud Platform

Product Applicability Guide for NIS2

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Background

In this Product Applicability Guide (PAG), we provide an evaluation of VMware products that make up and support the VMware Telco Cloud Platform, and how they may support Network and Information Systems Directive 2 (NIS2) requirements. These products virtualize and abstract the physical technology layers such as compute, storage, and network, the essence of Telco Cloud Platform. The evolving technology landscape that is modernizing the data center is also modernizing the virtual desktop environment and mobile device management while making inroads to consolidate and automate Information Technology (IT) resources. Broadcom prioritizes data protection and system security features within the Telco Cloud Platform. Broadcom developed a framework that incorporates Telco Cloud Platform product capabilities aligned to NIS2 requirements.

The NIS2 Directive establishes a set of cybersecurity obligations for essential and important entities across the European Union (EU). It requires organizations to implement risk-based technical and organizational measures to ensure a high level of cybersecurity. The PAG offers a unique perspective on how VMware products can support these obligations by helping organizations meet risk-management and reporting requirements outlined in NIS2.

Broadcom engaged Tevora, an independent third-party IT audit firm, to conduct a review of the Telco Cloud Platform solution alignment to NIS2. This document is the culmination of Tevora's discussions with Broadcom product teams to perform a thorough evaluation of VMware product capabilities mapped to NIS2 requirements.

Tevora is a leading security consulting firm specializing in enterprise risk, compliance, information security solutions, and threat research. Tevora offers a comprehensive portfolio of information security solutions and services to clients in virtually all industries. This PAG will navigate readers through the NIS2 Directive and highlight applicable VMware product capabilities.

VMware Telco Cloud Platform and NIS2

Today's infrastructures are heterogeneous in nature, built upon collaborations between internally constructed products and third-party sourced components, all guided by a customer's complex business and compliance requirements.

Broadcom approaches compliance with a view that understands the complexity in environments and addresses those areas where virtualization can be leveraged to develop a more secure environment. This focused view on compliance is reflected in the VMware Compliance Solutions framework, which allows for wide-ranging adoption of regulatory controls.

The phrase "security by design" identifies architectural decisions and default settings inside VMware products that are integrated into the product lifecycle. This approach reflects the process Broadcom follows to weave in security through all stages of the product lifecycle, and not as an afterthought. A compliance-capable design follows the philosophy that mapping Telco Cloud Platform solution capabilities to NIS2 security requirements can result in a solution that has been vetted as compliance capable. This overlap between products and compliance requirements establishes a new level marrying security and non-security product capabilities to also achieve operational innovation.

What is VMware Telco Cloud Platform?

VMware Telco Cloud Platform refers to a cloud-based platform specifically designed for telecommunications service providers (CSPs), offering advanced features for managing and automating virtualized network functions (VNFs) and cloud-native network functions (CNFs) with capabilities like unified automation, orchestration, and multi-domain assurance.

What is NIS2?

The NIS2 Directive is an EU-wide cybersecurity regulation adopted in 2022 to strengthen the overall level of cybersecurity across Member States. It replaces the original NIS Directive and expands the scope of entities that must meet enhanced cybersecurity obligations. NIS2 establishes a regulatory framework designed to protect essential and important services across sectors such as energy, healthcare, transportation, digital infrastructure, manufacturing, financial services, and ICT service management.

Member States are responsible for enforcing NIS2, conducting supervisory activities, and ensuring that organizations comply with the Directive's requirements. Ultimately, NIS2 provides a unified foundation for cybersecurity across the EU and sets expectations for consistent protections and accountability.

Revision History

Date	Rev	Author	Comments
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Solution Area	Telco Cloud Infrastructure Tier Key Products
Orchestration	Telco Cloud Automation
Infrastructure as a Service (IaaS)	VMware ESX™, VMware vCenter®, VMware vSAN™, VMware NSX®
Containers as a Service (CaaS)	Tanzu Kubernetes Grid

Disclaimer (Tevora)

The opinions stated in this guide concerning the applicability of VMware ® products to the NIS2 requirements are the opinions of Tevora. All readers are advised to perform individual product evaluations based on organizational needs.

Disclaimer (Broadcom)

This document is intended to provide general guidance for organizations that are considering VMware solutions to help them address compliance requirements. The information contained in this document is for educational and informational purposes only. This document is not intended to provide regulatory advice and is provided “AS IS”. Broadcom makes no claims, promises, or guarantees about the accuracy, completeness, or adequacy of the information contained herein. Organizations should engage appropriate legal, business, technical, and audit expertise within their specific organization for review of regulatory compliance requirements.

Executive Summary

Introduction

What is NIS2?

The NIS2 Directive is the EU's primary cybersecurity regulation to strengthen the overall resilience of essential and important entities that provide critical services across Member States. NIS2 expands the scope of sectors covered under the original NIS Directive and establishes cybersecurity, risk-management, governance, and incident-reporting requirements across the EU. It aims to ensure that organizations implement appropriate and proportionate technical, operational, and organizational measures to manage cybersecurity risks and prevent or minimize the impact of incidents that disrupt critical services.

NIS2 applies to organizations operating in sectors such as energy, healthcare, transportation, manufacturing, financial services, digital infrastructure, public administration, and ICT service management. The Directive introduces more prescriptive expectations for risk management, supply chain security, business continuity, vulnerability handling, secure system design, and oversight from executive management. Member States are responsible for national implementation, supervision, and enforcement through designated competent authorities. Providers of electronic communication networks are considered part of the digital infrastructure sector under Annex I of NIS2.

Essential and Important Entities Under NIS2

NIS2 categorizes regulated organizations into two groups: Essential Entities and Important Entities. This classification determines the level of regulatory oversight, supervisory expectations, and enforcement actions applied under the Directive.

Essential Entities

Essential Entities are organizations that provide services whose disruption would have significant societal, economic, or security impact across the EU. These entities are subject to the highest level of regulatory supervision, including both reactive and proactive oversight. Essential Entities must implement cybersecurity measures that are aligned with Article 21 and are subject to increased supervisory scrutiny, including audits, inspections, and mandatory risk-management documentation.

Essential Entities typically operate in sectors such as:

- Energy
- Transportation
- Banking and Financial Market Infrastructures
- Healthcare
- Drinking Water and Wastewater
- Digital Infrastructure and Cloud Computing (Includes Telecommunication Sector)
- Public Administration
- Space

Important Entities

Important Entities are organizations that provide critical or economically significant services, but whose disruptions have comparatively lower systemic risk. Although they must meet the same cybersecurity and risk-management obligations as Essential Entities, their supervisory regime is primarily reactive, meaning enforcement is triggered when deficiencies, incidents, or noncompliance are identified.

Important Entities typically operate in sectors such as:

- Postal and Courier Services
- Waste Management
- Manufacturing of Critical Products (e.g., Medical Devices, Electronics, Chemicals)
- Digital Providers (e.g., Online Marketplaces, Search Engines, Social Networking Platforms)
- Food Production and Distribution
- Research Organizations

How does NIS2 Work?

Rather than prescribing a catalog of controls, NIS2 establishes a set of risk-based cybersecurity measures, governance measures, and reporting requirements that organizations must implement according to the nature of their services, operational risk exposure, and societal impact. These measures are outlined within Chapter IV Cybersecurity Risk Management Measures and Reporting Obligations and include:

- Article 20: Governance
- Article 21: Cybersecurity Risk Management Measures
    a) Policies on risk analysis and information system security
    b) Incident handling
    c) Business continuity, such as backup management and disaster recovery, and crisis management
    d) Supply chain security, including security-related aspects concerning the relationships between each entity and its direct suppliers or service providers
    e) Security in network and information systems acquisition, development and maintenance, including vulnerability handling and disclosure
    f) Policies and procedures to assess the effectiveness of cybersecurity risk-management measures
    g) Basic cyber hygiene practices and cybersecurity training
    h) Policies and procedures regarding the use of cryptography and, where appropriate, encryption
    i) Human resources security, access control policies and asset management
    j) The use of multi-factor authentication or continuous authentication solutions, secured voice, video and text communication systems within the entity, where appropriate
- Article 22: Union Level Coordinated Security Risk Assessments of Critical Supply Chains
- Article 23: Reporting Obligations

NIS2 establishes a holistic, organization-wide risk-management approach. NIS2 also introduces stronger governance and accountability requirements, including obligations for senior management to oversee cybersecurity, approve risk-management measures, and face potential penalties for non-compliance. Not all security measures apply equally to all organizations. Requirements depend on sector, size, and operational impact. Essential and important entities must ensure that their cybersecurity programs align with the Directive's expectations and reflect best practices relevant to their industry.

As the regulatory environment continues to evolve across Europe, aligning security practices with NIS2 provides organizations a foundation for demonstrating compliance and supporting cross-border operations.

Scope and Approach

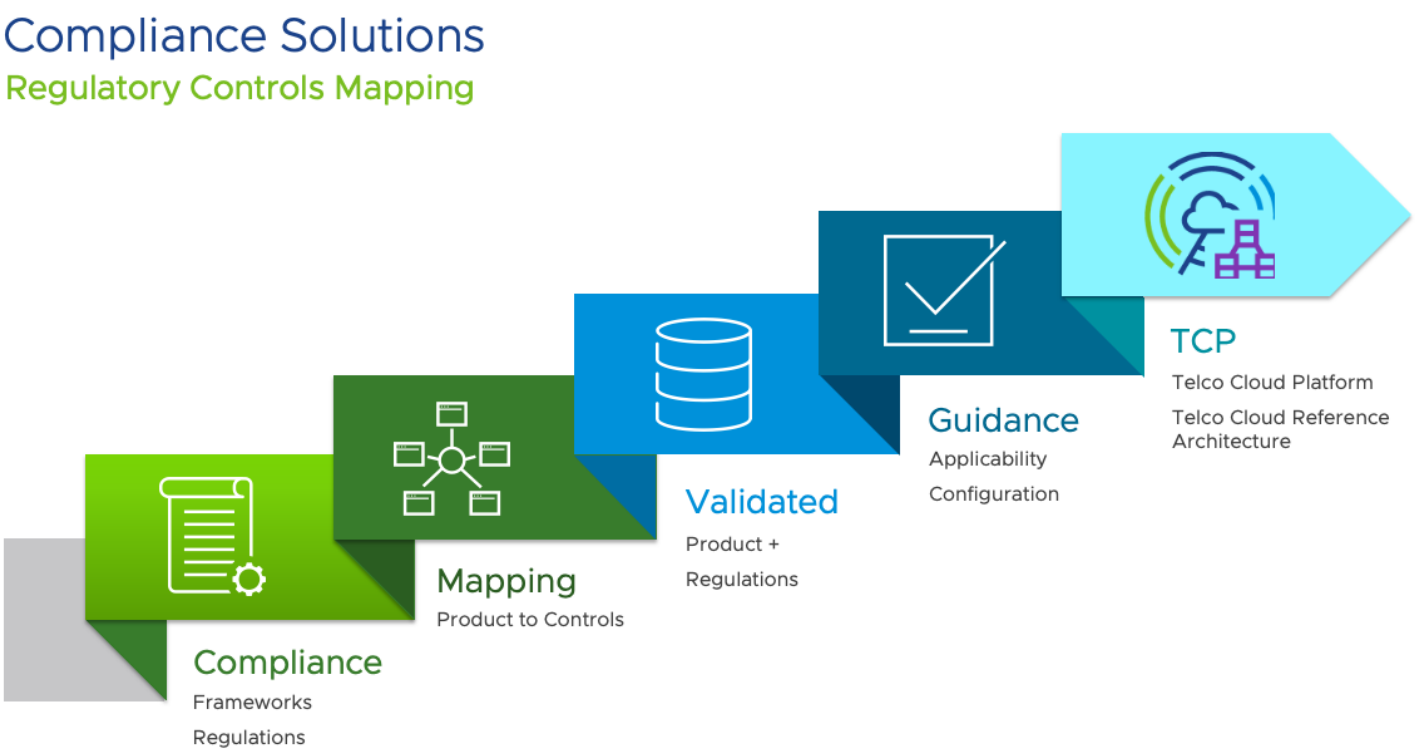
The Telco Cloud Platform solution covers a wide number of products and architectures. The platform and each of its component products contain features that could be mapped to some NIS2 requirements. This guide expands to account for Telco Cloud Infrastructure Tier components underneath the Telco Cloud Platform umbrella. Telco Cloud Infrastructure Tier provides the virtualization run-time environment with network functions and resource isolation for VM and container workloads. Virtualized compute, storage, and networking are delivered as an integrated solution through VMware vSphere®, VMware vSAN™, and VMware NSX®. The infrastructure tier is the lowest tier of the Telco Cloud Platform. It is aimed at VM-based Virtual Network Functions (VNFs) and is delivered through a Virtual Infrastructure Manager (VIM) layer. The infrastructure tier is optimized for telco-class workloads to enable the delivery of quality and resilient services. Infrastructure high availability, performance, and scale considerations that are built into this tier form the foundation for upper-level tiers. The scope of this guide is limited to those requirements supported either technically or through direct API integration. Additional technologies required in addition to VMware products are not identified.

Our Approach

This PAG is intended to provide information for all security and compliance practitioners on Tevora's recommended usage of the VMware technical stack to address regulatory compliance obligations and enhance the security of their services through the security and compliance of NIS2 requirements. It is up to each organization to identify the applicable NIS2 requirements that are in scope. Organizations must assess their exposure, sector-specific obligations, and operational dependencies to identify the appropriate and proportionate cybersecurity measures required under the Directive. The PAG focuses on capabilities of the Telco Cloud Platform solution at the requirements-category level, as each organization will need to identify its scope, conduct its own risk assessment, and determine the relevance and applicability of each Article 21 requirement to its business objectives. As a result, the level of implementation and supporting technical measures may vary from one organization to another based on factors such as criticality, operational impact, and sector-specific risks.

Appendix B outlines specific product capabilities for Telco Cloud Platform and their alignment to NIS2 requirements.

In addition to the NIS2 requirements, we used eleven (11) security lenses that serve as a baseline to evaluate the Telco Cloud Platform solution. From the ground up, Broadcom strives to design, define, and deliver compliance solutions to customers. The compliance solution begins with a compliance context (e.g., requirements from the appropriate standards in question). Next, the technical requirements applicable to the VMware products are mapped to in-scope compliance requirements. Finally, an independent audit evaluation of the design is conducted. The output is a solution that has interwoven compliance requirements into the end solution available to customers. Below is an overview of this process



Outside of the process described above, these eleven (11) areas are broad categories of controls that are implemented within today's security programs. They can be used to further understand the broader technology concepts used to build security architectures and to implement controls to mitigate risks.

The eleven (11) security lenses include:

- Automated Security
- System Hardening
- Compliance Validation
- System Access
- Data Segmentation
- System Monitoring
- Data Encryption & Protection
- Network Protection
- Endpoint Protection
- Trusted Execution/Secure Boot
- Software Development Lifecycle

Evaluating the Telco Cloud Platform through the additional layer of security lenses helps security and compliance practitioners understand how products deliver the features required not only to support compliance with NIS2 requirements but also to comport with general security best practices.

Tevora reviewed the high-level product design, followed by a detailed examination of data flows, features, architectures, and capabilities across all in-scope products to identify applicable controls. The testing considered all potential configurations that allow Telco Cloud Platform components to support each requirement.

The evaluation produced this guide to provide executives, technology experts, and security and compliance practitioners with insights to enhance security and compliance postures using VMware products. The flexibility of Telco Cloud Platform in feature deployment allows for connection with preexisting systems to further fortify security, privacy, and compliance. Understanding this flexibility is key to understanding how VMware products can be deployed with continuous compliance in mind.

In-Scope VMware Product List

VMware Telco Cloud Platform is a purpose-built, carrier-grade cloud services platform that enables Communications Service Providers (CSPs) to modernize their networks and rapidly deploy 4G, 5G, and edge network functions (NFs). It is a unified platform that supports both virtual network functions (VNFs) and cloud-native network functions (CNFs) on a consistent, horizontal infrastructure.

The platform is an integrated solution combining the foundational, high-performance VMware infrastructure layer with a multi-domain orchestration and automation solution that accelerates the design, deployment, and lifecycle management of network functions and services across the network. Embedded within the platform, this component allows CSPs to build, manage, and run containerized workloads (CNFs) using a CNCF-conformant Kubernetes distribution, expedite the innovation cycle to deliver new applications and services, reduce operational complexities, and achieve substantial TCO savings over other approaches, further accelerating CSPs' cloud modernization journey to 5G.

The Telco Cloud Infrastructure Tier is the foundation for the Telco Cloud Platform. For detailed information refer to Telco Cloud Platform infrastructure components.

The infrastructure tier focuses on the following major objectives:

- Provide a platform for creating a horizontal, disaggregated platform, supporting hardware and network functions from any vendor.
- Provide Software-Defined Telco Cloud Infrastructure
- Provide Software-Defined Storage
- Provide Software-Defined Networking

The following lists the components associated with Telco Cloud Platform that are in-scope for this Product Applicability Guide (PAG)

- VMware ESX™ – ESX is a purpose-built bare-metal hypervisor that installs directly onto a physical server. With direct access to and control of underlying resources, ESX is more efficient than

hosted architectures and can effectively partition hardware to increase consolidation ratios and cut costs for our customers.

- VMware vCenter® – vCenter provides centralized management of virtual infrastructure. IT administrators can bolster security and availability, simplify day-to-day tasks, and reduce the complexity of managing virtual infrastructure.
- VMware vSAN™ – vSAN is a core building block for Telco Cloud Platform, delivering enterprise-class, flash-optimized, and secure storage for all of the user's critical workloads.
- VMware NSX -- NSX is a network virtualization platform which creates, deletes, and restores software-based virtual networks. With network virtualization, the functional equivalent of a network hypervisor reproduces the complete set of Layer 2 through Layer 7 networking services (for example, switching, routing, access control, firewalling, QoS) in software.
- VMware Tanzu Kubernetes Grid – Tanzu Kubernetes Grid provides a Kubernetes platform to run containerized network functions on the Telco Cloud Platform.
- Telco Cloud Automation – Telco Cloud Automation automates the deployment and management of network functions and services within the Telco Cloud Platform.

The infrastructure components of Telco Cloud Platform, their interactions with each other, and how they meet CSP requirements are described in the Telco Cloud Reference Architecture Guide.

Overview of Broadcom and NIS2 Requirement Mapping

Best Practice Area (Lens)	NIS2 Articles	Capability Description	VMware Product Applicability
Automated Security	n/a	Automated Deployment, Automated Remediation	• n/a
Data Segmentation	n/a	Network & Host Firewall, Information Flow	• NSX
System Hardening	Article 20 Article 21	Configuration Management, Patch Management, Vulnerability Management	• ESX • Tanzu Kubernetes Grid • Telco Cloud Automation
Compliance Validation	Article 20 Article 21 Article 22	Configuration Management	• n/a
System Access	Article 21	Two-Factor Authentication, Identity and Access Management	• vCenter • ESX • Tanzu Kubernetes Grid
System Monitoring	Article 20 Article 21 Article 22 Article 23 Article 29	Security Information Event Monitoring (SIEM), Database Monitoring	• vCenter • Tanzu Kubernetes Grid
Data Encryption & Protection	Article 21 Article 23 Article 29	Data at Rest Encryption, Data in Motion Encryption, System Backup & Restore	• vSAN
Network Protection	Article 21 Article 23 Article 29	Intrusion Prevention System, Web Application Firewall	• NSX
Endpoint Protection	Article 21	Endpoint A/V and Malware Prevention, File Integrity Monitoring, Data Leakage Protection, Mobile Device Management	• NSX • ESX
Trusted Execution/Secure Boot	n/a	Execution Integrity	• NSX • ESX
Software Development Lifecycle (SDLC)	Article 21	Configuration Integrity	• Telco Cloud Platform

VMware Control Capabilities

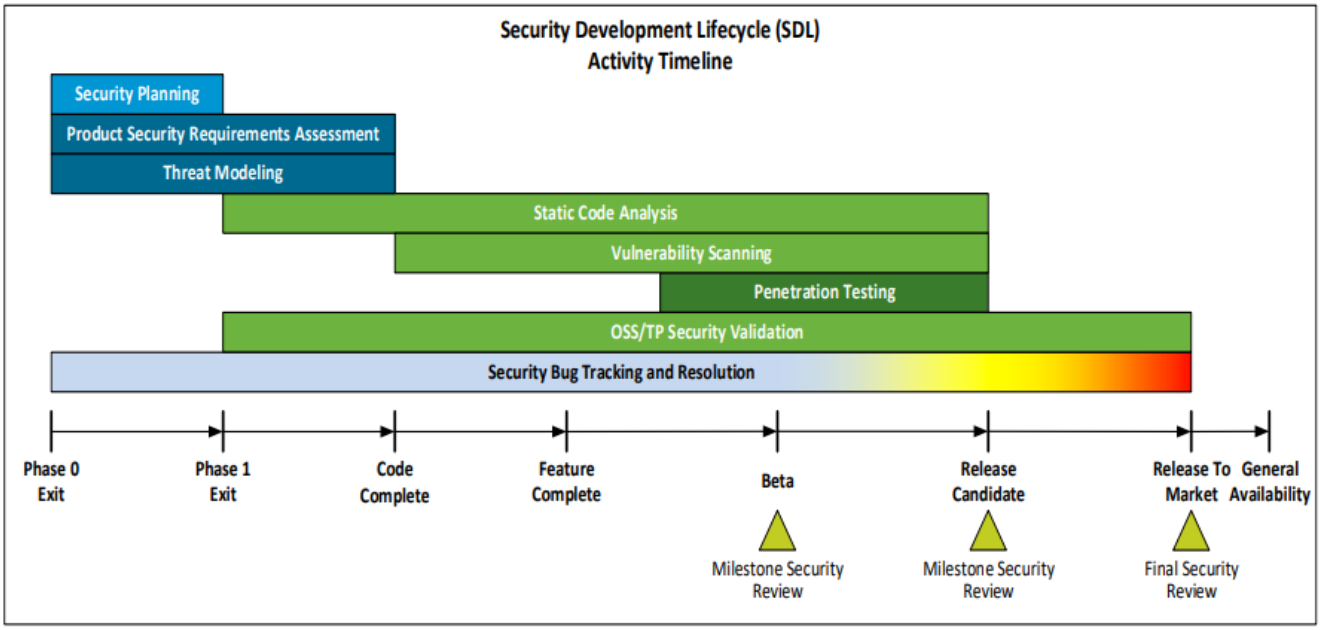
VMware Validated Design and Software Development Process

Broadcom has developed the VMware Telco Cloud Platform™ as a carrier-grade cloud services platform to help Communication Service Providers to expedite the innovation cycle to deliver new NFV applications and services using a reference architecture design that is validated and provides the details required to confidently deploy the solution. The Telco Cloud Reference Architecture Guide is available to anyone and is published on Broadcom website.

VMware Security Development Lifecycle (SDL) designs security in all phases of Telco Cloud Platform. This principled approach to designing security, overseen by VMware Product Security from the start, is important for supporting NIS2 compliance, as the Directive requires that security be integrated into underlying technical architectures and reinforced through appropriate organizational and administrative policies.

With compliance and security, Broadcom improves the quality of its products and solution platforms that can support organizations using NIS2 requirements.

As further reference to the primary purposes of a control family, each detail segment provides the applicable security lens defined within the Broadcom approach. These lenses are hallmarks of a mature security program addressing common areas of vulnerabilities.



Governance and Technical Security Requirements

NIS2 does not use control families or classifications such as those found in NIST frameworks. Instead, the Directive defines risk-management requirements and reporting obligations under Chapter IV Cybersecurity Risk Management Measures and Reporting Obligations. To improve clarity within this PAG, Tevora grouped these requirements into five categories:

- Governance and Accountability
- Risk Management and Technical/Organizational Measures
- Incident Response Expectations and Reporting timelines
- Supply Chain and Third-Party Risk Management
- Compliance and Supervisory Enforcement Mechanisms

Several NIS2 requirements involve establishing and maintaining documented policies, governance procedures, communication workflows, and personnel responsibilities. Although these obligations are not fulfilled through technology alone, VMware products play a key supporting role to provide insights that help organizations implement and enforce their administrative measures.

VMware Governance and Accountability

Article 20: Governance

This article establishes requirements for member states to ensure that essential and important entities require senior management to approve the risk management program and documentation as well as outline the management structure within risk management. Other aspects include assigning ownership of risks to senior management or Board of Directors and including risk management into their security awareness program. All Telco Cloud Platform components provide secure, resilient, and scalable infrastructure with built in monitoring, compliance controls, and redundancy that help identify, mitigate, and respond to risks.

Applicable Security Lens:

- System Hardening
- System Monitoring
- Compliance Validation

VMware Risk Management and Technical/Organizational Measures

Article 21: Cybersecurity Risk Management Measures

This article establishes the risk-based cybersecurity measures that an organization should implement to manage cybersecurity risks and protect information systems. The article covers a variety of technical, operational, and organizational measures such as policy implementation, risk management, supply chain security, authentication, encryption, training, incident handling, secure communications, software development, and vulnerability management.

Applicable Security Lens:

- System Hardening
- Compliance Validation
- System Access
- Data Encryption and Protection
- Software Development Lifecycle (SDLC)
- Endpoint Protection
- Network Protection

Applicable VMware Product(s):

- ESX
- vCenter
- vSAN
- NSX
- Tanzu Kubernetes Grid
- Telco Cloud Automation

VMware Product Capabilities

Business Continuity and Recovery in accordance with NIS2 will primarily be governed by administrative policy. VMware products such as vCenter can support the need for data backups as detailed within the organization's policy when properly used. Backups of vCenter can be made at a data and configuration level versus traditional storage/image-based backups. This reduces recovery time by letting the system administrator (SA) spin up a new vCenter while simultaneously importing the backed-up data.

Further, for components related to system recovery, NSX has been designed to support high availability. VMware Avi Load Balancer can be configured, at the organization's discretion, to automatically distribute workloads across multiple servers to prevent any one server from becoming overloaded. This is to ensure uptime as required by SLAs or other operating needs.

The Broadcom approach to security extends into its development process. While constant iteration is a priority, security is interwoven into every stage from ideation and design to development and production. Static code analysis, security, and privacy considerations at the design phase run through multiple levels of approval, in addition to performance-level assessments. Developers participate in extensive secure code training and regularly attend working sessions in collaboration with security compliance and privacy teams to stay abreast of evolving trends and vulnerabilities.

VMware security programs and practices establish requirements “by design” to evolve methodologies of protection against new “in-the-wild threats”. This takes effect throughout the development process and is developed into products. Products are tested by first-class vulnerability scans and penetration tests prior to any full release or version update.

Broadcom monitors all security alerts that can affect VMware products security and issues a security patch if needed. When configured appropriately, products such as vCenter assist in the updates and patching for complementary VMware products, e.g., ESX.

NSX supports multi-tenancy enabling isolation of security and networking configurations across tenants. NSX users across tenants can access the same NSX Manager to provision networking and security objects on shared hosts. On the ESX host, IP-based storage traffic and vMotion traffic can be isolated from other traffic to further protect them from attack.

All port groups on standard switches can be configured to reject forged transmits, guest MAC address changes and guest promiscuous mode requests. NSX Manager enforces approved authorizations for controlling the flow of management information within the network device based on information flow control policies.

NSX and its micro-segmentation can then enforce defined information flow guidelines. NSX Manager™ can sync with RBAC to restrict access based on specified group names. The information contained within these data flows are then secured at rest with the vSAN Encryption feature in vSAN. ESX further segments processes within resource pools.

Telco Cloud Platform requires the use of AES256 cryptographic protocols. The vSAN Encryption feature in vSAN certifies that all data stored within a customer’s Telco Cloud Platform environment is encrypted to industry standards.

Telco Cloud Platform Reference Architecture Guide forces security requirements to maintain adequate levels of encryption, and logging. All infrastructure tier components of the Telco Cloud Platform solution include digital signatures and cryptographically strong hash functions.

vCenter has the capability to restrict cryptographic operations such as key management server (KMS) functions and encrypting/decrypting virtual machine disks to a specific cryptographic role. This prevents all administrators from having access to cryptographic operations adding additional granularity into access rights.

The Photon operating system provides a minimal Linux distribution that supports FIPS validated cryptographic modules. The operating system supports kernel hardening, process isolation, and SSH key authentication to prevent unauthorized access to system resources.

As for access control, VMware products have third-party integration capabilities to allow organizations to integrate with external identity providers that support multi-factor authentication (MFA) and single-sign-on (SSO) to strengthen authentication needs and restrict access. Organizations can also integrate their Active Directory (AD)/Lightweight Directory Access Protocol (LDAP) through use of VMware Application Programmable Interfaces (APIs) to refine access at all levels of their virtual stack.

For all products within the Telco Cloud Platform, Access Control can be implemented at a granular level. This is presented through Role-Based Access Control (RBAC) mechanisms natively available. User management interfaces are provided to control password complexity and user profiles and to access review tasks.

Additional product capability details can be found at VMware Compliance IDs listed in Appendix B.

VMware Incident Response Expectations and Reporting Timelines

Article 23: Reporting Obligations

This article establishes mandatory reporting timelines and requirements for incidents to facilitate information sharing and ensure affected parties are notified. Incident information is required to be compiled into reports to communicate details of the incident including investigation results and remediation activity. Telco Cloud Platform infrastructure tier components can assist in the information exchange and auditing of those events attributed to technical elements through the integration of a SIEM. Across all products, Broadcom employs secure information-sharing processes.

Applicable Security Lens:

- System Monitoring
- Network Protection
- Data Encryption & Protection

Applicable VMware Product(s):

- ESX
- vCenter
- NSX
- Tanzu Kubernetes Grid
- Telco Cloud Automation

VMware Product Capabilities

Across all products, rich logging features exist to allow administrators to ascertain who logged in, the origin, at what time (coordinated through NTP), and whether the attempt was a success or failure.

Logs can be pointed to third-party management tools through API integration if desired. ESX can be configured to support a persistent log location for locally stored logs. vCenter can be configured for specific-day retention for logs and events. ESX affords administrators the ability to adjust the richness and frequency of audit logs. Tanzu Kubernetes Grid allows notifications to be configured for sending alerts in the event of a processing failure or storage exceeding capacity. Logs can be configured across all products to be sent to a central log server.

To support non-repudiation, administrators are advised to design strong access control surrounding Administrator passwords. All administrative actions should be logged and reviewed on a consistent basis.

For advancing an organizational audit process, it is recommended that a Security Incident Event Management (SIEM) platform be coordinated with VMware products to ingest logs.

Additional product capability details can be found at VMware Compliance IDs listed in *Appendix B*.

Article 29: Cybersecurity Information-Sharing Arrangements

This article requires information sharing practices and agreements to be established, implemented and maintained so organizations can voluntarily exchange cybersecurity information among themselves and with Member States. Telco Cloud Platform infrastructure tier components can assist in the information exchange and auditing of those events attributed to technical elements through the integration of a SIEM. Across all products, Broadcom employs secure information-sharing processes.

Applicable Security Lens:

- System Monitoring
- Network Protection
- Data Encryption & Protection

VMware Supply Chain and Third-Party Risk Management

Article 21: Cybersecurity Risk Management Measures

Supplier and Third-Party Risk Management Measures

Regarding supply chain and third-party risk management, this article outlines various measures requiring entities to assess the security of third-party suppliers and services while determining and documenting risk responses for any risks third parties pose. The article also requires contracts containing security requirements and provisions for supply chain members to comply with organization security requirements.

While VMware products do not have capabilities that would directly help an organization meet these requirements, these products are designed with compliance and security in mind during all phases of SDLC . This can provide organizations assurance that these products can help meet their compliance and security needs. To support this, Broadcom provides security configuration and architecture documentation for organizations to securely design the Telco Cloud Platform.

Broadcom also highly recommends not installing or using any software not supported by the VMware products to minimize the threat to the organization's infrastructure.

Applicable Security Lens:

- System Hardening
- Compliance Validation
- Data Encryption and Protection

Article 22: Union Level Coordinated Security Assessments of Critical Supply Chains

This article requires member states to participate in coordinated EU supply-chain risk assessments and identify critical ICT services, systems and products. Although this article does not specifically outline requirements for entities, entities are expected to support and comply with outcomes of this assessment. This can include providing relevant information on suppliers and dependencies as well as implementing mitigation measures for any suppliers or services deemed high-risk. There are no direct capabilities that VMware products provide for these requirements. However, VMware products provide secure, resilient, and scalable infrastructure with built in monitoring, compliance controls, and redundancy that help identify, mitigate, and respond to risks.

Applicable Security Lens:

- System Monitoring
- Compliance Validation

VMware Compliance and Supervisory Enforcement Mechanisms

The supervision and enforcement chapter of the NIS2 Directive sets out how member states monitor compliance and apply corrective measures to ensure both essential and important entities are meeting the requirements as outlined within the directive. Authorities are granted broad powers to request information and evidence, conduct on-site and off-site checks, perform security scans and audits, order remediation actions, and require the specific implementation of risk management measures. Although this is a regulatory requirement, VMware products provide secure, resilient, and scalable infrastructure with built in monitoring, compliance controls, and redundancy that organizations can align with their security documentation to support NIS2 compliance.

Conclusion

To meet evolving regulatory needs, security programs now must define applicable controls at early stages.

The considerations that Broadcom brings to bear on continuous compliance for clients comes from its development culture, which constructs requirements that balance functionality and security for all deployable products. These policies provide customers with the confidence to include the Telco Cloud Platform solution suite within their architecture and overall NIS2-aligned risk-management program.

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Appendix A: NIS2 Requirements Mapping

VMware Product	NIS2 Requirements Supported
Telco Cloud Platform	Article 21
VMware vCenter	Article 21, Article 23
VMware ESX	Article 21, Article 23
VMware vSAN	Article 21
VMware NSX	Article 21, Article 23
VMware Tanzu Kubernetes Grid	Article 21, Article 23
Telco Cloud Automation	Article 21, Article 23

Appendix B: Telco Cloud Platform Product Capability Relationship with NIS2

vSAN

Product	Capability ID	Product Capability	NIS2 Requirement
vSAN	vSAN_001	vSAN capacity should be verified with a virtual infrastructure administrator to ensure capacity is sufficient and planned appropriately for normal operations of the Telco Cloud Platform.	Article 21 2(e)
	vSAN_002	The vSAN Health Check is able to download the HCL from Broadcom to check compliance against the underlying vSAN Cluster hosts. To ensure the vCenter server is not directly downloading content from the internet, this functionality must be disabled. If this feature is necessary, an external proxy server must be configured.	Article 21 2(e)

Telco Cloud Platform

Product	Capability ID	Product Capability	NIS2 Requirement
Telco Cloud Platform	TCP _001	Telco Cloud Platform can be backed up and restored with an image-based solution or a file-based solution.	Article 21 2(c)
	TCP _002	An authoritative time source can be configured to synchronize system clocks, allowing the correlation of events that occur across multiple systems.	Article 21 2(b)
	TCP _003	Security patches and updates can be installed for Telco Cloud Platform	Article 21 2(e)

	TCP _004	Telco Cloud Platform should use approved Class 3 and Class 4 SSL certificates for software signing and business-to-business transactions.	Article 21 2(h) Article 21 2(j)
	TCP _005	Telco Cloud Platform can be configured to not be exposed directly to the internet.	VMware Best Practice
	TCP _006	Access control policies (e.g., identity-based policies, role-based policies, attribute-based policies) and access enforcement mechanisms (e.g., access control lists, access control matrices, cryptography) can be employed on the application server to support the principle of least privilege for users and service accounts.	Article 21 2(i)
	TCP_007	Telco Cloud Platform supports FIPS mode that only supports secure cryptographic algorithms. This option is only available for new installations and the settings you apply during bring-up are used for future upgrades. You cannot change the FIPS security mode setting after bring-up.	Article 21 2(h) Article 21 2(j)
	TCP_008	Telco Cloud Platform supports multifactor authentication for both privileged and non-privileged accounts through an integration with MFA supported OIDC or LDAP identity providers.	Article 21 2(i)

NSX

Product	Capability ID	Product Capability	NIS2 Requirement
NSX	NSX_001	NSX system clocks can be synchronized to NTP to enable accurate and universal time source logging.	Article 21 2(b)
	NSX_002	Virtual network tunnel traffic can be isolated from other traffic either through a VLAN segment or by physical isolation.	Article 21 2(j)
	NSX_003	Only trusted hypervisors should be used on the logical network data plane.	Article 21 2(j)
	NSX_004	Access permissions to NSX Manager Virtual Appliance can be restricted to intended administrators.	Article 21 2(i)
	NSX_005	NSX can be used to verify that IPv4 DNS is authorized and secured.	Article 21 2(j)
	NSX_006	NSX backup directory can be stored on a SFTP server with read or write access limited to a specific backup user.	Article 21 2(c)
	NSX_007	NSX Manager clusters can have rules applied to it that separate the nodes onto different physical hosts.	VMware Best Practice
	NSX_008	NSX Manager can be used to download patches or upgrades for the NSX. After installation the MD5/SHA1 output can be validated with the MD5/SHA1 sum values on the Broadcom secure website.	Article 21 2(e)
	NSX_009	NSX comes with pre-defined roles that can be assigned to enable Role Based Access Control. In addition, new roles can be developed based on an inventory of functionalities to be used as custom roles. This capability can support both Separation of Duties, as well as the concept of Least Privilege.	Article 21 2(i)

	NSX_010	NSX Distributed Firewall supports anti-spoofing measures to detect and block outbound IP packets that contain illegitimate packet attributes.	VMware Best Practice
	NSX_011	NSX has the capability to redirect logs to a SIEM or copy logs and send them to another logging tool for analysis.	Article 21 2(b)
	NSX_012	NSX Manager can be configured to integrate with VMware Cloud Foundation® Identity Broker. This configuration supports non-repudiation of actions taken by an administrator.	Article 21 2(i)
	NSX_013	NSX Manager supports device session timeout at the end of the session or after an organization-defined period of timeout.	Article 21 2(i)
	NSX_014	NSX Manager supports an organization-defined API account lockout duration after repeated failed login attempts	Article 21 2(i)
	NSX_015	NSX Manager supports an API account lockout policy to limit the number of failed login attempts before locking out the user.	Article 21 2(i)
	NSX_016	NSX Manager supports an account lockout policy to limit the number of failed login attempts before locking out the user.	Article 21 2(i)
	NSX_017	NSX Manager can be configured to capture audit records to establish, correlate and investigate events relating to an incident	Article 21 2(b) Article 23
	NSX_018	NSX Manager can be used to configure the minimum password length policy. The default value is 12 characters.	VMware Best Practice
	NSX_019	Multiple NSX Manager nodes are deployed with VIP or external load balancers to preserve network element state information. This helps facilitate continuous network operations with minimal to no disruption.	VMware Best Practice

	NSX_020	Client API rate limits, client API concurrency limits and Global API concurrency limits can be configured in NSX Manager to protect the system from denial-of-service (DoS) attacks.	VMware Best Practice
	NSX_021	It is highly recommended to not install/use any software not supported by Broadcom to minimize the threat to infrastructure. Do not add other software components to the NSX appliances as it is an untested configuration and could potentially interfere with the operation of the security functions they provide.	Article 21 3
	NSX_022	NSX patches or upgrades can be installed as soon as the update bundles are available.	Article 21 2(e)
	NSX_023	Storage networks can be isolated from any other network to protect against rouge VM attacks.	VMware Best Practice
	NSX_024	The NSX Manager system clock can be configured so time stamps for audit records are mapped to Coordinated Universal Time (UTC).	Article 21 2(b)
	NSX_025	NSX Manager obtains its public key certificates from approved certificate authorities. NSX Manager uses a certificate for each manager and one for the cluster VIP. In some cases, these are the same, but each node and cluster VIP certificate must be checked individually.	Article 21 2(j)
	NSX_026	NSX Manager can be configured to conduct backups on an organizational defined schedule.	Article 21 2(c)
	NSX_027	NSX Manager can be configured to disable SSH by default which will prevent users from bypassing NSX RBAC- Access Control and audit controls. SSH can be temporarily turned on when needed.	Article 21 2(j)
	NSX_028	The backup-users password can be configured to meet the organization's defined password requirements.	VMware Best Practice

	NSX_029	A valid TLS certificate or a self-signed certificate can be used for the communication between NSX Manager and VMware Cloud Foundation® Identity Broker.	Article 21 2(j)
	NSX_030	NSX Manager supports an organization-defined CLI account lockout duration after repeated failed login attempts	Article 21 2(i)
	NSX_031	NSX Manager supports an CLI account lockout policy to limit the number of failed login attempts before locking out the user.	Article 21 2(i)
	NSX_032	NSX Manager supports CLI session timeout at the end of the session or after an organization-defined period of inactivity.	Article 21 2(i)
	NSX_033	NSX Manager can be configured to disable unused local accounts.	VMware Best Practice
	NSX_034	NSX supports TLS 1.2 and is the default protocol.	Article 21 2(h) Article 21 2(j)
	NSX_035	NSX Manager supports SNMP - Media Protection v3 for commercial-grade security, including authentication, authorization, access control, and privacy.	VMware Best Practice
	NSX_036	NSX must be updated to the current version to support the latest security updates.	VMware Best Practice
	NSX_037	NSX Manager provides management plane protection for denial-of-service attack by having API rate limiting to limit the number of transactions per second and concurrent transactions to the NSX REST API. However, as of now, there is no built-in mechanism to restrict the usage of REST APIs. But such usage can be monitored from logs and ensure that the API usage is granted and approved by your local site policy.	VMware Best Practice
	NSX_038	SFTP is supported to encrypt backups and restorations for NSX.	Article 21 2(c) Article 21 2(h)

	NSX_039	The SFTP server can be hardened with general best practices and guidelines to protect the backups and restorations from common threats and vulnerabilities.	Article 21 2(b)
	NSX_040	A remote syslog server or SIEM can be enabled to collect NSX logs. It is important to ensure the Syslog server is authorized and secure within the NSX Manager Virtual Appliance.	Article 21 2(b)
	NSX_041	NSX Manager can be used to verify TLS is being used within the LDAP server settings.	Article 21 2(i)
	NSX_042	NSX Distributed Firewall can be configured to send traffic log entries to a central audit server for management and configuration of the traffic log entries.	Article 21 2(b)
	NSX_043	The NSX Manager has the capability to monitor possible port scan attacks by reviewing the log entries or employing a log management product.	VMware Best Practice
	NSX_044	NSX supports multifactor authentication for both privileged and non-privileged accounts through an integration with MFA supported OIDC or LDAP identity providers.	Article 21 2(i)

ESX

Product	Capability ID	Product Capability	NIS2 Requirement
ESX	ESX_001	Remote access to ESX via SSH can be controlled on the ESX host.	VMware Best Practice
	ESX_002	Host based authentication to ESX via SSH can be disallowed on the ESX host.	VMware Best Practice
	ESX_003	Root logins to the ESX host can be disabled.	VMware Best Practice
	ESX_004	Remote logins to the ESX host can be configured to reject authentication using an empty password.	VMware Best Practice

	ESX_005	User environment settings can be disabled for the ESX host to prevent users from potentially bypassing access restrictions.	VMware Best Practice
	ESX_006	The ESX host SSH daemon can be configured with strict mode to prevent users from logging into the system as another user.	VMware Best Practice
	ESX_007	The ESX host can be configured to disallow compression.	VMware Best Practice
	ESX_008	The ESX host can be configured to block gateway ports to prevent the SSH tunnel from accessing other systems on the internal network, circumventing firewalls and network access control lists.	Article 21 2(j)
	ESX_009	The ESX can be configured to disable x11 forwarding over SSH.	VMware Best Practice
	ESX_010	The ESX host can be configured to block network tunnels over an SSH connection.	Article 21 2(j)
	ESX_011	A session timeout can be configured for the SSH client session by setting the client alive message count.	Article 21 2(i)
	ESX_012	A session timeout can be configured for clients that do not actively respond during the connection.	Article 21 2(i)
	ESX_013	The ESX host enables password enforcement rules such as password complexity.	VMware Best Practice
	ESX_014	The ESX host firewalls can be configured to only allow traffic from the ESX management network.	VMware Best Practice
	ESX_015	The vSphere Client will present a warning if SSH or the ESX Shell is enabled on the ESX hosts.	VMware Best Practice
	ESX_016	Lockdown mode can be configured on the ESX host disabling direct access. To do so the host will need to be managed remotely from a vCenter Server.	VMware Best Practice
	ESX_017	The ESX host supports an account lockout policy to limit the number of failed login attempts before locking out the user.	Article 21 2(i)

	ESX_018	The ESX host supports session timeout at the end of the session or after an organization-defined period of inactivity.	Article 21 2(i)
	ESX_019	Bridge Protocol Data Unit filtering can be enabled on the ESX host to automatically drop any BPDU packets from being sent to the physical switch.	VMware Best Practice
	ESX_020	The ESX host firewalls rules can be configured to deny all traffic by default.	Article 21 2(j)
	ESX_021	The ESX host firewalls rules can be configured to block all outbound network traffic.	Article 21 2(j)
	ESX_022	The ESX host can enable a password reuse restriction setting to prohibit password reuse.	VMware Best Practice
	ESX_023	The ESX host supports the ability to disable SSH to prevent users from remotely connecting to the ESX shell. SSH can be temporarily enabled for maintenance.	Article 21 2(j)
	ESX_024	The ESX host supports the ability to disable the ESX Shell Service preventing users from bypassing vCenter RBAC-Access Control. The ESX Shell Service can be temporarily enabled for maintenance.	VMware Best Practice
	ESX_025	In support of User Access Controls and in particular process controls such as adding, modifying, or removing users, ESX hosts can be integrated with Active Directory to support access controls.	Article 21 2(i)
	ESX_026	The ESX hosts support the configuration of a logon banner before granting DCUI access. This provides the user with a privacy and security notice consistent with organizational policies, laws and regulations.	Article 21 2(i)

	ESX_027	The ESX hosts support the configuration of a logon banner before granting SSH access. This provides the user with a privacy and security notice consistent with organizational policies, laws and regulations.	Article 21 2(i)
	ESX_028	An exceptions list for lockdown mode can be configured. These predetermined users will not lose their permissions when the host enters lockdown mode.	VMware Best Practice
	ESX_029	Security patches and updates can be installed for ESX hosts.	Article 21 2(e)
	ESX_030	ESX hosts can be configured to send all locally stored logs to a persistent log location.	Article 21 2(b)
	ESX_031	When adding ESX hosts to Active Directory a default group can be configured. It is important to ensure the user/group is not added to the "ESX Admin" group by default.	Article 21 2(i)
	ESX_032	The ESX host has the capability to disable Inter-VM transparent page sharing.	VMware Best Practice
	ESX_033	ESX system clocks can be synchronized to an NTP time server to enable accurate and universal time source logging.	Article 21 2(b)
	ESX_034	The ESX host NTP Service Startup Policy can be configured to start and stop with the host.	Article 21 2(b)
	ESX_035	The ESX Image Profile should be verified to only allow signed vSphere Installation Bundles (VIB) The ESX host can be configured to support signed VIBs preventing unsigned bundles from being installed.	VMware Best Practice
	ESX_036	All port groups on ESX standard switches can be configured to reject forged transmits.	Article 21 2(j)
	ESX_037	All port groups on ESX standard switches can be configured to reject guest MACaddress changes.	Article 21 2(j)

	ESX_038	All port groups on ESX standard switches can be configured to reject guest promiscuous mode requests.	Article 21 2(j)
	ESX_039	ESX hosts support a DCUI Access List to allow specific predetermined users to exit lockdown mode in case a host becomes isolated from vCenter.	VMware Best Practice
	ESX_040	ESX hosts can be configured to send logs to a centralized log server	Article 21 2(b)
	ESX_041	The ESX host can be configured to enforce an unlock timeout after a user account is locked out.	Article 21 2(i)
	ESX_042	The ESX host supports the ability to disable the Managed Object Browser (MOB). The MOB is disabled by default since it does not support access controls and could be used to obtain information about a host being targeted for unauthorized access.	VMware Best Practice
	ESX_043	ESX supports DCUI session timeout at the end of the session or after an organization-defined period of timeout.	VMware Best Practice
	ESX_044	The ESX host provides the ability to restrict dvfilter network API's. If the dvfilter network API is not in use, then it should not be configured to send network information to a virtual machine.	Article 21 2(j)
	ESX_045	The ESX host can be configured to isolate network traffic to a dedicated VLAN. It is highly recommended that vMotion traffic is isolated from all other traffic.	Article 21 2(j)
	ESX_046	ESX Hosts can be configured to capture audit records to establish, correlate, and investigate events leading up to an outage or attack.	Article 21 2(b)
	ESX_047	ESX supports host client session timeout at the end of the session or after an organization-defined period of timeout.	VMware Best Practice

	ESX_048	OpenSSH on the ESX host supports a FIPS 140-2 validated cryptography module and is enabled by default.	Article 21 2(h)
	ESX_049	The ESX host can be configured to isolate network traffic to a dedicated VLAN. It is highly recommended that IP-based storage traffic is isolated from all other traffic.	Article 21 2(j)
	ESX_050	ESX hosts support bidirectional CHAP authentication for traffic.	Article 21 2(j)
	ESX_051	All port groups on standard switches can be configured to be different from their native virtual local area network value.	VMware Best Practice
	ESX_052	The virtual Local Area Network value of all port groups on standard switches can be configured. They should not be configured to VLAN 4095 unless Virtual Guest Tagging (VGT) is required.	VMware Best Practice
	ESX_053	All port groups on standard switches can be configured but they should not be configured to VLAN values reserved by upstream physical switches.	VMware Best Practice
	ESX_054	CIM service account permissions on the ESX host can be limited to a read-only or custom role.	VMware Best Practice
	ESX_055	The ESX host supports secure TLS (TLS 1.2) cryptographic controls for all endpoints.	Article 21 2(h) Article 21 2(j)
	ESX_056	ESX can be configured to enable Secure Boot for all guest OS. With UEFI Secure Boot enabled, a host refuses to load any UEFI driver or app unless the operating system bootloader has a valid digital signature.	VMware Best Practice
	ESX_057	The ESX Shell can be configured to disable port forwarding for ESX host SSH daemon.	VMware Best Practice
	ESX_058	The OpenSLP service can be disabled on the ESX host and be configured to start and stop manually when needed.	VMware Best Practice

	ESX_059	The default self-signed host certificate issued by the VMware Certificate Authority (VMCA - Assessment, Authorization and Monitoring) can and should be replaced with a DOD-approved certificate when the host is accessed directly.	Article 21 2(j)
	ESX_060	The ESX SNMP - Media Protection service can be hardened if it is in use by configuring the proper trap destination. If the service is not in use, it can be disabled.	VMware Best Practice
	ESX_061	When sending syslog data to a remote host, ESX can be configured to use any combination of TCP - Contingency Planning, UDP and SSL transports. When using SSL, the server certificate must be validated to ensure that the host is connecting to a valid syslog server.	Article 21 2(j)
	ESX_062	The ESX host supports a session timeout for the vSphere API at the end of the session or after an organization-defined period of timeout.	VMware Best Practice
	ESX_063	The ESX host http proxy daemon is capable of using FIPS 140-2 validated cryptographic modules to protect the confidentiality of remote access sessions.	Article 21 2(h) Article 21 2(j)
	ESX_064	The ESX host can be configured to not override virtual machine configurations. This can be managed and automated per-VM vmx settings.	VMware Best Practice
	ESX_065	The ESX host can be configured to not override virtual machine legger settings. This can be managed and automated per-VM vmx settings.	VMware Best Practice
	ESX_066	The ESX host can enable strict x509 verification for SSL syslog endpoints.	Article 21 2(h)
	ESX_067	Volatile key destruction can be enabled on the ESX host.	VMware Best Practice
	ESX_068	The ESX host enables password enforcement rules such as maximum password age.	VMware Best Practice

	ESX_069	The CIM service on ESX can be disabled if not in use.	VMware Best Practice
	ESX_070	The ESX host SSH daemon can be configured to only use FIPS 140-2 validated ciphers.	Article 21 2(h)
	ESX_071	The SNMP - Media Protection service on ESX can be disabled if not in use.	VMware Best Practice
	ESX_072	A trusted platform module (TPM) 2.0 can be installed and enabled on the ESX host	VMware Best Practice
	ESX_073	Broadcom provides the latest security updates on vendor support websites, including end of life status.	Article 21 2(e)
	ESX_074	IT infrastructure management interfaces can be isolated on their own network segment or as part of an isolated management network to support better access controls.	Article 21 2(j)
	ESX_075	Hardware management controllers and other hardware management functionality/tools can be disabled if they are not in use or provide little value.	VMware Best Practice
	ESX_076	Remote access to ESX via SSH can be controlled on the ESX host.	VMware Best Practice
	ESX_077	Host based authentication to ESX via SSH can be disallowed on the ESX host.	VMware Best Practice
	ESX_078	Root logins to the ESX host can be disabled.	VMware Best Practice
	ESX_079	Remote logins to the ESX host can be configured to reject authentication using an empty password.	VMware Best Practice
	ESX_080	ESX supports multifactor authentication for both privileged and non-privileged accounts through an integration with MFA supported OIDC or LDAP identity providers.	Article 21 2(i)

vCenter

Product	Capability ID	Product Capability	NIS2 Requirement
vCenter	vCenter_001	Parallel devices on the virtual machine that have not been authorized can be detected through manual review or by running a command in the Power CLI command prompt.	VMware Best Practice
	vCenter_002	Serial devices on the virtual machine that have not been authorized can be detected through manual review or by running a command in the Power CLI command prompt.	VMware Best Practice
	vCenter_003	USB devices on the virtual machine that have not been authorized can be detected through manual review or by running a command in the Power CLI command prompt.	VMware Best Practice
	vCenter_004	Shared salt values can be disabled on virtual machines.	VMware Best Practice
	vCenter_005	Copy operations can be disabled on virtual machines.	VMware Best Practice
	vCenter_006	Drag and drop operations on virtual machines can be disabled.	VMware Best Practice
	vCenter_007	Paste operations can be disabled on virtual machines.	VMware Best Practice
	vCenter_008	Virtual disk shrinking can be disabled on virtual machines.	VMware Best Practice
	vCenter_009	Virtual disk erasure can be disabled on virtual machines.	VMware Best Practice
	vCenter_010	Console connection sharing can be limited through the vSphere Client. This prevents a non-administrator in the VM from connecting to the	VMware Best Practice

		console and potentially observing an administrator's actions.	
	vCenter_011	Informational messages from the VM to the VMX file can be configured to a specific capacity on the virtual machine. It is recommended that its value is set to 1MB but can be modified for organizational needs.	VMware Best Practice
	vCenter_012	Within the vSphere Client a configuration can be made to prevent unauthorized device modifications in virtual machines by blocking non-admin users or processes from connecting, disconnecting, or adjusting device settings.	VMware Best Practice
	vCenter_013	Virtual machines can be configured to obtain host information from the hypervisor. It is recommended to disable this configuration for all VM's unless a particular VM requires this information for performance monitoring.	VMware Best Practice
	vCenter_014	Independent, non-persistent disks on the virtual machines should not be used and can be detected through manual review or by running a command in the Power CLI command prompt.	VMware Best Practice
	vCenter_015	Floppy disks on the virtual machines that have not been authorized can be detected by running a command in the Power CLI command prompt.	VMware Best Practice
	vCenter_016	CD/DVD devices on the virtual machines that have not been authorized can be detected through manual review or by running a command in the Power CLI command prompt.	VMware Best Practice
	vCenter_017	Service account reviews can be performed in the system. No application should share a service account when connecting to vCenter Server.	VMware Best Practice
	vCenter_018	vCenter can enable a password reuse restriction setting to prohibit password reuse	VMware Best Practice

	vCenter_019	ESX Hosts can be configured to capture audit records to establish, correlate, and investigate events leading up to an outage or attack.	Article 21 2(b)
	vCenter_020	vCenter can set the distributed port group to a Promiscuous Mode. This prevents all connected virtual machines from reading all packets across the network.	VMware Best Practice
	vCenter_021	The vCenter Server includes a vSphere Client extensibility framework, which provides the ability to extend the vSphere Client with menu selections or toolbar icons that provide access to vCenter Server add-on components or external, web-based functionality. vSphere Client plugins or extensions run at the same privilege level as the user. When using these plug-ins ensure only verified trusted sources are allowed.	VMware Best Practice
	vCenter_022	All distributed port groups on vCenter can be configured to reject MAC address changes.	VMware Best Practice
	vCenter_023	vCenter enables password enforcement rules such as a minimum upper case character policy.	VMware Best Practice
	vCenter_024	vCenter can manage excess capacity, bandwidth, or other redundancy to limit the effects of information-flooding types of Denial of Service (DoS) attacks by enabling Network I/O Control (NIOC).	Article 21 2(e)
	vCenter_025	vCenter enables password enforcement rules such as a minimum password length policy.	VMware Best Practice
	vCenter_026	vCenter enables password enforcement rules such as a minimum lower case character policy.	VMware Best Practice
	vCenter_027	vCenter comes with pre-defined roles that can be assigned to enable Role Based Access Control. In addition, new roles can be developed based on an inventory of functionality to be used as custom roles. This capability can support both Separation of Duties, as well as the concept of Least Privilege.	Article 21 2(i)

	vCenter_028	The distributed virtual switch can export NetFlow information about traffic crossing the switch. NetFlow exports are not encrypted and can contain information about the virtual network, making it easier for a man-in-the-middle attack to be executed successfully. If NetFlow export is required, verify that all NetFlow target Internet Protocols (IPs) are correct.	VMware Best Practice
	vCenter_029	The vSAN Health Check is able to download the public Hardware Compatibility list. However, it is important to ensure that the vCenter is not directly downloading content from the internet. It is recommended to have this functionality disabled and if this feature is necessary, an external proxy server can be configured instead.	Article 21 2(e)
	vCenter_030	The vSAN Datastore can be configured with a unique name preventing confusion if multiple datastores are in use.	VMware Best Practice
	vCenter_031	vCenter enables password enforcement rules such as a maximum password age policy.	VMware Best Practice
	vCenter_032	A session timeout can be configured for the management session after a certain period of inactivity.	VMware Best Practice
	vCenter_033	VPX users can have their minimum password length set separately. It is highly recommended that the default length is set to at least 32 characters for these accounts.	VMware Best Practice
	vCenter_034	By default, vCenter will change the "vpxuser" password automatically every 30 days. Ensure this setting meets site policies. If it does not, configure it to meet password aging policies.	VMware Best Practice
	vCenter_035	vCenter enables password enforcement rules/policies such as a minimum number of special characters policy.	VMware Best Practice

	vCenter_036	vCenter enables password enforcement rules/policies such as a minimum number of numeric characters policy.	VMware Best Practice
	vCenter_037	vCenter can be configured with an account lockout policy to set a time interval between failures.	Article 21 2(i)
	vCenter_038	vCenter can be configured with an account lockout policy to require an administrator to unlock accounts after excessive login failures.	Article 21 2(i)
	vCenter_039	vCenter can be configured with an account lockout policy to set a maximum number of failed login attempts.	VMware Best Practice
	vCenter_040	vCenter provides a built-in SSO administrator account. This account should be used for emergencies only and be governed by policy.	Article 21 2(i)
	vCenter_041	All distributed port groups on vCenter can be configured to reject forged transmits.	Article 21 2(j)
	vCenter_042	PCI passthrough can be enabled to direct assignment of hardware PCI functions to virtual machines. It is recommended to only have this feature enabled on virtual machines that need it.	VMware Best Practice
	vCenter_043	vCenter provides the capability to enable/disable the distributed virtual switch health check. It is turned off by default and is recommended to only be used when troubleshooting.	Article 21 2(e)
	vCenter_044	All port groups on standard switches can be configured to be different from their native virtual local area network value.	VMware Best Practice
	vCenter_045	The virtual Local Area Network value of all port groups on standard switches can be configured but they should not be configured to VLAN values reserved by upstream physical switches.	VMware Best Practice
	vCenter_046	vCenter is configured to use TLS 1.2 by default.	Article 21 2(h)

			Article 21 2(j)
	vCenter_047	The default self-signed, VMware Certificate Authority (VMCA - Assessment, Authorization and Monitoring)-issued vCenter reverse proxy certificate can be replaced with a DOD-approved certificate. The use of a DOD certificate on the vCenter reverse proxy and other services assures clients that the service they are connecting to is legitimate and trusted.	Article 21 2(j)
	vCenter_048	vCenter supports the configuration of a logon banner before granting access to the system. This provides the user with a privacy and security notice consistent with organizational policies, laws and regulations.	VMware Best Practice
	vCenter_049	Users/groups can be assigned to a default role within vCenter. It is important to ensure the user/group is not added to any of the 4 default roles that contain cryptographic permissions.	VMware Best Practice
	vCenter_050	Users/groups can be assigned specific cryptographic permissions within vCenter. It is important to ensure only the administrator is assigned any of these permissions.	VMware Best Practice
	vCenter_051	Mutual CHAP for vSAN Integrity targets can be configured from the vSphere client. When enabled, vSphere performs bidirectional authentication of both the Integrity target and host.	VMware Best Practice
	vCenter_052	Key Encryption Keys (KEKs) can be configured to be reissued at regular intervals for the vSAN encrypted databases.	VMware Best Practice
	vCenter_053	The dvFilter API can be used to filter and inspect network traffic. Access to the virtual machines through the dvFilter network APIs can be controlled and limited.	VMware Best Practice
	vCenter_054	vCenter provides the ability for users to be added to predefined or custom groups. vCenter SSO integrates with PAM in the underlying Photon	Article 21 2(i)

		operating system so members of the "SystemConfiguration.BashShellAdministrators" SSO group can log on to the operating system without needing a separate account. It is recommended that users have strict authorization to be a part of this group.	
	vCenter_055	vCenter provides the ability for users to be added to predefined or custom groups. The vSphere "TrustedAdmins" group grants additional rights to administer the vSphere Trust Authority feature. It is recommended that users have strict authorization to be a part of this group	Article 21 2(i)
	vCenter_056	vCenter can push near real time logs to be stored in an external centralized log server.	Article 21 2(b)
	vCenter_057	vCenter has the capability to configure alarms tied to event IDs to notify users if a certain event or action takes place. These alerts can be set up to alert the appropriate personnel about any SSO account actions.	Article 21 2(i)
	vCenter_058	Backups of vCenter can be made at a data and configuration level versus traditional storage/image-based backups. This reduces recovery time by letting the system administrator (SA - System and Services Acquisition) spin up a new vCenter while simultaneously importing the backed-up data.	Article 21 2(c)
	vCenter_059	vCenter can generate volumes of security relevant application level events and has the capability to send these events to be stored in an external centralized log server.	Article 21 2(b)
	vCenter_060	vCenter supports SNMP - Media Protection v3 security features where SNMP - Media Protection is required.	VMware Best Practice
	vCenter_061	vCenter can be configured to isolate network traffic to a dedicated VLAN or other network segments. It is highly recommended that IP-based storage traffic	Article 21 2(j)

		on distributed switches is isolated from all other traffic.	
	vCenter_062	vCenter task retention and vCenter event retention can be set to a defined number of days. The default value is 30 days but can be modified based on organizational needs.	Article 21 2(b)
	vCenter_063	vCenter allows port groups to be configured to VLAN Trunking, allowing the vSwitch to pass all network frames in the specified range to the attached virtual machines without modifying the virtual local area network (VLAN) tags. It is recommended that this is configured only if required and as authorized by Virtual Guest Tagging.	VMware Best Practice
	vCenter_064	vCenter can employ active directory as an authentication mechanism. Additionally, the identity provider can be embedded with either Windows Integrated Authentication or LDAP.	Article 21 2(i)
	vCenter_065	vCenter supports LDAPS to protect the confidentiality of LDAP communications.	Article 21 2(i)
	vCenter_066	The LDAP identity source can be configured using a limited privileged account.	Article 21 2(i)
	vCenter_067	The guest operating system can be configured to be locked when the last console user disconnects. This setting only applies to Windows-based virtual machines with VMware tools installed.	VMware Best Practice
	vCenter_068	Virtual machines can use 3D graphic features. These features are off by default and are recommended to not be used unless the virtual machine requires 3D functionality.	VMware Best Practice
	vCenter_069	vCenter supports 256-bit AES-GCM - Configuration Management encryption for vMotion on the virtual machine.	Article 21 2(b) Article 21 2(h)

	vCenter_070	vCenter supports the ability to use premade templates to deploy virtual machines.	VMware Best Practice
	vCenter_071	The virtual machine console can be removed from the standard virtual machine role.	VMware Best Practice
	vCenter_072	vCenter supports SNMP - Media Protection v3 commercial-grade security, including authentication, authorization, access control, and privacy. Previous versions of the protocol contained well-known security weaknesses that were easily exploited. Therefore, SNMP - Media Protection v1/2 receivers can be disabled, while SNMP - Media Protection v3 is configured in another control. vCenter exposes SNMP - Media Protection v1/2 in the UI and SNMP - Media Protection v3 in the CLI.	VMware Best Practice
	vCenter_073	The vCenter Native Key acts as a key provider for encryption-based capabilities such as encrypted virtual machines without requiring an external KMS solution. When enabling this feature a backup can be created using a strong password.	Article 21 2(h)
	vCenter_074	vCenter can be configured to capture logs throughout the system.	Article 21 2(b)
	vCenter_075	The virtual machines can be configured to capture logs. Log collection is enabled by default through the ESX hypervisor.	Article 21 2(b)
	vCenter_076	The ESX hypervisor maintains logs for each individual VM by default. These logs contain information including but not limited to power events, system failure information, tools status and activity, time sync, virtual hardware changes, vMotion migrations, and machine clones. By default, the size of these logs is unlimited, and they are only rotated on vMotion or power events. This can cause storage issues at scale for VMs that do not vMotion or power cycle often. It is	Article 21 2(b)

		recommended to set a log size appropriate for your system.	
	vCenter_077	By default, 10 of these logs for virtual machines are retained. This is normally sufficient for most environments, but this configuration can be verified, updated, and maintained.	Article 21 2(b)
	vCenter_078	vSphere Fault Tolerance performs frequent checks between a primary VM and secondary VM so the secondary VM can quickly resume from the last successful checkpoint. The checkpoint contains the VM state that has been modified since the previous checkpoint. Encryption can be enabled for Fault Tolerance. By default, the encryption is set to "Opportunistic", which means it enables encryption only if both the primary and secondary host are capable of encryption.	Article 21 2(h)
	vCenter_079	vCenter can automatically synchronize internal information system clocks to an authoritative time server.	Article 21 2(b)
	vCenter_080	Unnecessary services or services not in use can be disabled such as CDP or LLDP network discovery protocols.	VMware Best Practice
	vCenter_081	The vSphere Distributed Virtual Switch can enable port mirroring sessions allowing traffic to be mirrored from one source to a destination. To maintain security, any unauthorized port mirroring sessions on distributed switches should be promptly removed.	VMware Best Practice
	vCenter_082	vCenter is intended to be managed through VAMI, vSphere Client, and APIs. SSH can be disabled unless and temporarily turned on for troubleshooting and support.	Article 21 2(j)

	vCenter_083	vCenter network firewall rules in conjunction with network/perimeter firewall rules can be configured to deny all traffic, except explicitly designated traffic.	VMware Best Practice
	vCenter_084	Unnecessary computer interfaces such as NIC's can be disabled through vCenter.	VMware Best Practice
	vCenter_085	Security patches and updates can be installed for vCenter. vCenter should be updated first and then users can proceed with updating ESX if an update is available.	Article 21 2(e)
	vCenter_086	vCenter supports multifactor authentication for both privileged and non-privileged accounts through an integration with MFA supported OIDC or LDAP identity providers.	Article 21 2(i)

Telco Cloud Automation

Product	Capability ID	Product Capability	NIS2 Requirement
Telco Cloud Automation	TCA_1	A session timeout can be configured for Telco Cloud Automation main UI. This timeout is configured within the administration settings, and the timeout is also applied to the API.	Article 21 2(i)
	TCA_2	Three types of system level user accounts are created during the deployment of Telco Cloud Automation Manager or Telco Cloud Automation Control Plane.	VMware Best Practice
	TCA_3	System level accounts do not have default passwords configured. The passwords for each type of account can be set to ensure they meet organizations password requirements including complexity.	Article 21 2(i)

	TCA_4	System level accounts can be configured with password expiration. By default, admin user accounts expire every 90 days.	Article 21 2(i)
	TCA_5	By default, the Telco Cloud Automation appliance is configured to lock system level accounts after three failed login attempts within a 15-minute period.	Article 21 2(i)
	TCA_6	For remote connections, all Telco Cloud Automation appliances include the SSH protocol.	VMware Best Practice
	TCA_7	The Web Console for the Telco Cloud Automation Manager or Telco Cloud Automation Control Plane supports the ability to disable SSH to prevent users from remotely connecting to the appliances. SSH can be temporarily enabled for maintenance and troubleshooting.	Article 21 2(i)
	TCA_8	A log bundle can be requested for Telco Cloud Automation. The log bundle can contain application logs, database dump and Kubernetes cluster logs.	Article 21 2(b)
	TCA_9	Audit logs for the Telco Cloud Automation can be monitored within the Telco Cloud Automation UI. This UI contains all events that occurred within the Telco Cloud Automation platform for two months.	Article 21 2(b)
	TCA_10	NTP servers can be configured to synchronize system clocks, allowing the correlation of events that occur across multiple systems.	Article 21 2(b)
	TCA_11	All non-essential ports and protocols can be deactivated. The minimum incoming and outgoing ports for Telco Cloud Automation can be configured.	VMware Best Practice
	TCA_12	Broadcom documents the minimum required ports and protocols for Telco Cloud Automation, allowing firewalls and routers to be easily configured.	VMware Best Practice

	TCA_13	Software installation can be managed through monitoring the inventory of installed products.	VMware Best Practice
	TCA_14	Software installations can be managed and restricted to essential software. It is not recommended to use any third-party software that Broadcom does not support.	VMware Best Practice
	TCA_15	Telco Cloud Automation keeps track of its installation, patching, and upgrade history to audit version changes.	Article 21 2(e)
	TCA_16	Backup bundles can be created through the Telco Cloud Automation Appliance Manager. When a backup is generated, there is a togglable option to encrypt the backup.	Article 21 2(c)
	TCA_17	Within the appliance manager, you can configure an SFTP protocol setting to automatically transfer backup files to a designated remote server.	Article 21 2(c)
	TCA_18	Broadcom provides security advisories for any vulnerabilities or guidance related to VMware products. The installation, patching and upgrade history can be used to verify the released advisories are enforced.	Article 21 2(e)
	TCA_19	VMware vSphere environment can be hardened to provide a secure Telco Cloud Automation environment. Broadcom provides vSphere hardening standards.	VMware Best Practice
	TCA_20	Users must harden the software infrastructure components such as the operating system, supporting software and database software to ensure a completely hardened and secure environment.	VMware Best Practice
	TCA_21	Telco Cloud Automation and other HTTPS endpoints can be secured using signed certificates.	Article 21 2(j)

	TCA_22	The network communication settings for any VMware virtual appliances and host machines are available to review and configure.	VMware Best Practice
	TCA_23	By default, the Telco Cloud Automation Manager and Telco Cloud Automation Control Plane appliances are deployed with self-signed TLS certificates. These default certificates can be replaced with organization-issued or trusted CA-signed certificates to ensure secure, authenticated communications.	Article 21 2(j) Article 21 2(h)
	TCA_24	Before installing Telco Cloud Automation, the integrity of the installation media can be verified by comparing the checksum sha256 value with the sha256 hash published on Broadcom Support Portal.	Article 21 2(e)
	TCA_25	Telco Cloud Automation integrates Photon OS security patches and critical package updates into each release. Regularly upgrading to the latest Telco Cloud Automation release ensures timely application of all underlying Linux security updates.	Article 21 2(e)
	TCA_26	Telco Cloud Automation can be deployed in an air gapped or internet-restricted environment, reducing exposure to external networks.	VMware Best Practice
	TCA_27	Telco Cloud Automation supports multifactor authentication for both privileged and non-privileged accounts through an integration with MFA supported OIDC or LDAP identity providers.	Article 21 2(i)

Tanzu Kubernetes Grid

Product	Capability ID	Product Capability	NIS2 Requirement
Tanzu Kubernetes Grid	TKG_1	Audit logs can be enabled on a standalone management cluster or a workload cluster before the cluster is deployed.	Article 21 2(b)

	TKG_2	Kubernetes audit logs can be forwarded to a specific log destination by using a log forwarder or customizing the settings in the audit log configuration	Article 21 2(b)
	TKG_3	Audit policy files can be created to finetune the information that is logged to ensure the system can capture the level of audit log detail defined by the organization.	Article 21 2(b)
	TKG_4	The audit log retention for the Kubernetes API can be configured.	Article 21 2(b)
	TKG_5	Customers can configure a remote server for offloading audit logs at a predefined frequency. The Photon operating system can offload real-time logs through the fluentbit package.	Article 21 2(b)
	TKG_6	Notifications can be configured to send an alert via syslog in the event of an audit processing failure or storage exceeding the predefined capacity threshold.	Article 21 2(b)
	TKG_7	NTP servers can be configured to synchronize system clocks, allowing the correlation of events that occur across multiple systems. If Tanzu Kubernetes Grid is running on vSphere, the hosts should be synchronized to the same NTP servers.	Article 21 2(b)
	TKG_8	Audit tools can and should be cryptographically signed so the organization has the capability of identifying when audit tools have been modified, manipulated, or replaced.	Article 21 2(b)
	TKG_9	Audit logs can be enabled on the Photon OS.	Article 21 2(b)
	TKG_10	RBAC can be configured for accounts that need access to Audit logs.	Article 21 2(i)

	TKG_11	Audit logging can be enabled to capture logs during Photon OS startup. This ensures all startup processes are captured.	Article 21 2(b)
	TKG_12	By default, non-privileged users are blocked from downloading and executing software.	VMware Best Practice
	TKG_13	The Photon OS can be configured to enable the tdnf package management tool which will cryptographically verify the authenticity of all software packages during installation.	VMware Best Practice
	TKG_14	The owner of each Kubernetes component (kubelet, manifest, API server, etcd, admin.conf, kubeadm.conf, kube proxy etc.) configuration files can be configured to limit access to the root to prevent unauthorized access.	VMware Best Practice
	TKG_15	The permissions on the different Kubernetes component (kubelet, manifest, API server, etcd, admin.conf, kubeadm.conf, kube proxy etc.) configuration files can be restricted to prevent an attacker from modifying Kubernetes in a way for them to bypass user authorizations	VMware Best Practice
	TKG_16	The Kubernetes kubelet can be configured to deny hostname override. This prevents a user from changing the hostname which would cause issues with matching the audit logs and metrics to the machine's actual hostname.	VMware Best Practice
	TKG_17	Unnecessary kernel modules on the Photon OS can be disabled so the operating system only provides essential capabilities.	VMware Best Practice
	TKG_18	The Kubernetes Controller Manager can be disabled to prevent the exposure of Kubernetes architecture details. This service can be enabled if it is deemed necessary.	VMware Best Practice
	TKG_19	The Kubernetes API server ports, protocols and services can be configured and disabled to provide only essential connections.	VMware Best Practice

	TKG_20	The Photon OS can be configured to disable the system from sending and accepting IPv4 Internet Control Message redirects.	VMware Best Practice
	TKG_21	An external OIDC or LDAP identity provider can be configured within the Tanzu Kubernetes Grid to manage accounts, identifiers, passwords, banners and messages while controlling user access to the workload clusters.	Article 21 2(i)
	TKG_22	Tanzu Kubernetes Grid supports multifactor authentication for network access to both privileged and non-privileged accounts through an integration with MFA supported OIDC or LDAP identity providers.	Article 21 2(i)
	TKG_23	Role Based Access Control can be configured for non-admin accounts within Tanzu Kubernetes Grid. In addition, new roles can be developed based on an inventory of functionality to be used as custom roles. This capability can support both Separation of Duties, as well as the concept of Least Privilege.	Article 21 2(i)
	TKG_24	A DoD login banner can be configured on any Linux based custom machine, including Tanzu Kubernetes Grid Photon OS nodes.	Article 21 2(i)
	TKG_25	Kubernetes audit logs can capture all requests made to the API server, including actions such as RoleBinding creation, modification, and deletion.	Article 21 2(b)
	TKG_26	Shared or group accounts are not configurable on Tanzu Kubernetes Grid.	VMware Best Practice
	TKG_27	The number of containers that run with higher privileges can be configured to a handful of namespaces to prevent privilege escalation.	VMware Best Practice
	TKG_28	Session timeout can be configured for the Photon OS.	Article 21 2(b)

	TKG_29	OpenSSH on the Photon OS supports FIPS 140-2 validated OpenSSL cryptography module.	Article 21 2(h)
	TKG_30	Photon OS can be configured to prohibit empty password logins even if a user somehow has an empty password on the OS.	Article 21 2(b)
	TKG_31	Approved Message Authentication Codes (MAC's) can be implemented on the Photon OS to protect the integrity of remote access sessions.	VMware Best Practice
	TKG_32	Tanzu Kubernetes Grid supports individual authentication prior to gaining access to or interacting with shared service accounts.	VMware Best Practice
	TKG_33	SSH access to Photon OS nodes uses key-based authentication.	VMware Best Practice
	TKG_34	The ESX host can be configured to enable bidirectional Challenge-Handshake Authentication Protocol for both the iSCSI target and host.	VMware Best Practice
	TKG_35	Password policies can be configured for local accounts on the Photon Operating System. The password configurations support complexity, length, minimum/maximum password age, reuse restrictions, and minimum number of characters changed.	Article 21 2(i)
	TKG_36	Photon OS supports the ability to require a password change upon first login.	VMware Best Practice
	TKG_37	The Photon OS supports reauthentication for privilege escalation.	VMware Best Practice
	TKG_38	Kubernetes namespaces can be configured to logically partition a single Kubernetes cluster into multiple sub-clusters. This can be used to separate user functionality from system management functionality.	VMware Best Practice
	TKG_39	Access to the kernel message buffer in the Photon OS can be restricted to prevent attackers from gaining additional system information as a nonprivileged user.	VMware Best Practice

	TKG_40	TCP syncookies can be configured on the Photon OS to verify the initiator is attempting a valid connection and not a flood source. This helps protect against denial of service (DOS) attacks stemming from a TCP SYN flood attack.	VMware Best Practice
	TKG_41	Secure Shell (SSH) sessions within the Photon OS can be configured to terminate after a predefined timeframe.	VMware Best Practice
	TKG_42	FIPS 140-2 validated cryptography can be enabled for the Photon OS	Article 21 2(h)
	TKG_43	Address space layout randomization can be implemented on the Photon OS to protect its memory from unauthorized code execution.	VMware Best Practice

About Tevora

Tevora is a leading management consulting firm specializing in enterprise risk, compliance, information security solutions, and threat research. We offer a comprehensive portfolio of information security solutions and services to clients in virtually all industries and also serve institutional and government clients.

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